S5-1

Pile driving noise and substrate vibrations impact hearing abilities of summer flounder (*Paralichthys dentatus*)

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The summer flounder (Paralichthys dentatus) is a bottom-dwelling flatfish native to the U.S. Atlantic coast. This species is important for the recreational and commercial fishing industry, generating $30 million (USD) annually. Lacking a swim bladder, flounders rely on particle acceleration rather than sound pressure for hearing. Their habitat overlaps with offshore windfarm sites employing pile-driving during construction, making flounders vulnerable to the complex field of impulsive noise and substrate vibrations generated during pile-driving. Data on how pile-driving noise and substrate vibrations impacts hearing sensitivity in benthic fish such as flounders are lacking. This study employed the auditory evoked potential methods (AEPs) to 1) determine the basic hearing range for the flounder, and 2) assess AEP-thresholds at 150 Hz before and after exposure to a real-world pile-driving setup. Flounder auditory sensitivity was measured after one 15-minute sequence of impact pile-driving and five 15-minute sequences of impact pile-driving; sensitivities were retested after a 3-week recovery period. Flounders exposed to pile-driving showed threshold shifts of 4–7 dB after just one 15-minute sequence of continuous pile-driving. Flounders exposed to five sequences showed a similar reduction in hearing sensitivity. After the 3-week recovery, mean hearing sensitivities improved for both treatments and were similar to control thresholds. These data suggest that the hearing abilities of flatfish, such as the summer flounder, may be compromised after relatively brief exposure to pile-driving events. This is particularly concerning given their benthic lifestyle and the distinct noise-response behaviors they exhibit compared to pelagic fish species.

S5-2

Is the high occurrence of loud, broadband, super-cavitation transients from ship propellers a problem for toothed whales?

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Vessel noise is the overwhelmingly dominant source of anthropogenic noise pollution at sea. It peaks below 200 Hz, many octaves below the best hearing frequencies of echolocating toothed whales. Accordingly, toothed whales are often considered a lowimpact group for vessel noise, although noise from cavitating propellers extends into the ultrasonic range. Here we provide evidence that fast propellers produce loud, ultrasonic pulses that we coined super-cavitation transients. To quantify these, we deployed an 8 m linear steel array of 4 hydrophones between November and December 2024 in the middle of Great Belt, Denmark, a highly trafficked route for commercial vessels. We recorded vessel passages, with a focus on localizing cavitation transients to derive their source properties. To find the noise contribution from each ship category, RMS levels at long range were also computed, when possible. Combining the AIS tracking data and the decidecade levels from the recordings, we were able to track and identify more than 200 vessel passages in a 100 m radius around the array, of which 28% were bulk carriers. Localized cavitation transients resulted in centroid frequencies (fc) of 51 kHz [43-71] with high source levels (SLpp) of 201 dB re 1 μPa (1 m) [199-202] and sound exposure levels (SEL10dB) of 145 dB re 1 $\mu Pa^2 s$ [143-146], unweighted. Based on previously reported thresholds, we argue that toothed whales could possibly be significantly impacted by the contribution from such super-cavitation noise, both in terms of hearing and behavioral response.

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S5-3

Characterizing vessel noise in Glacier bay National Park to inform soundscape management strategies

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Vessels are a pervasive source of anthropogenic noise that fragments the natural soundscapes in Glacier Bay National Park, Alaska, USA. More than 90% of human visitors to this largely marine Park travel by vessel, and many areas are accessible only by vessel. To effectively balance Park use against noise impacts and preserve natural soundscapes and quietude, a collaboration was formed between the U.S. National Park Service (NPS) and Cornell University to create an acoustic propagation modeling interface for NPS managers to quantify the effects of vessel speed regulation strategies on the underwater and above water soundscapes in the Park. The first steps towards this goal entail 1) characterizing source levels of vessel categories present in the park, 2) investigating the relationship between source level and transit speed by vessel category. In 2024, acoustic data were collected from various vessel types in the Park in accordance with the ANSI-ASA S12.64-2009 standard for the measurement of underwater noise from ships. Source level estimates were computed for cruise vessels, tour vessels, charter, vessels, private vessels, and NPS vessels; relationships between speed and source level were characterized for each vessel category. Both were compared to literature values. The next phase will be developing a program which will incorporate these source level estimates into simulations to calculate noise metrics at key underwater and above water locations throughout the park under various vessel traffic scenarios. These efforts will provide quantitative insights to support natural soundscape management by the NPS.

S5-4

Impacts of vessel noise to the acoustic behaviour of the Indo-pacific humpback dolphins in Northwest peninsular Malaysia

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Research on Indo-Pacific humpback dolphins (Sousa chinensis) in the Langkawi Archipelago, northwestern Peninsular Malaysia, has focused on visual observations since 2010, causing a significant knowledge gap regarding their acoustic ecology. Langkawi is one of Malaysia's top tourism sites and these coastal dolphins share their habitat with intensive maritime traffic and fishing activities. This study aims to investigate how boat noise impacts the dolphins' acoustic behaviour.

Between 2019 and 2024, 189 sightings were observed and over 180 hours of dolphin acoustic data were collected. A total of approximately 4,355 whistles were detected with durations and frequency modulations extracted to be compared to different types of group behaviours and group sizes. Whistle frequencies ranged from 1687.0 Hz to 37687.5 Hz with durations between 0.025 s to 1.9 s. Whistle rates were compared to the baseline levels of boat absence and during boat presence. Preliminary results showed significant differences in whistle rate whereby it was lower after boat presence compared to before. The higher whistle rates before boat presence may serve as a function to keep the group cohesion or to maintain group behaviour. Results showed that in smaller group sizes, whistle duration was longer after boat presence. Longer and more frequently emitted whistles could serve as additional information for communication in boat presence. These findings highlight the importance of incorporating acoustic monitoring into research and conservation strategies and is essential for developing effective mitigation action plans on the impacts of vessel noise on marine mammals in Malaysian waters.

S5-5

The reaction of bottlenose dolphins to brief, steady-state startle sounds in the wild

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Oceanic delphinids in areas of frequent anthropogenic operations are regularly exposed to intense noise. It has been demonstrated that high noise doses can create large responses in these animals. However, less is known about the effects of lower doses. Recent work on the startle reflex in cetaceans has highlighted the potential for behavioural change at comparatively low exposure levels. After demonstrating the existence of the startle reflex in captive bottlenose dolphins, we studied its effect on animal movement in the wild. We exposed groups of bottlenose dolphins to five brief (200 ms duration) startle sounds (10-20 kHz peak frequency) at irregular intervals over a 1 min period (max. received level 147 dB re 1 µPa) and compared their subsequent movement to the pre-exposure period and to control periods without noise exposure. During exposure, dolphins changed their heading and increased their swim speed in an avoidance response that lasted up to 4 min, resulting in a distance of 500-800 m from the source at the end of this period. In control periods, no change of behaviour was observed. Our results demonstrate a clear movement response to a relatively minor noise dose but also highlight its transient nature. While a short-term displacement will have little impact on animals and can be used to manage marine mammal occurence in areas that could otherwise lead to animal injury (e.g. fisheries, construction sites), startle sounds are ubiquitous in the anthropogenic soundscape. Further studies are needed to understand their broader effects on marine mammals.

S5-6

Using bioacoustic tools to reveal the effects of noise on songbirds and their automated detection

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Anthropogenic noise pollution is a growing threat to biodiversity, negatively affecting countless species. Acoustic monitoring has emerged as a transformative tool for assessing biodiversity and the impacts of noise on wildlife. In two successive studies, we investigated the effects of noise on songbird populations by deploying a "Phantom Turbine" and a “Phantom road”: broadcasting noise at a site without actual turbines or roads. The experiments followed three stages: 'before', 'noise-treatment', and 'after'. We monitored birds’ abundance using mist-netting and through passive acoustic monitoring of freely flying birds while measuring background noise levels. In both studies, bird abundance decreased during noise treatment. It is then essential to ensure that reductions in vocalizations are not due to impaired detection -we addressed this in two ways: 1) using the hurdle model to explore changes in vocalization in relation to background noise levels and 2) comparing BirdNet recall and precision across different experimental phases. This verification is important also for projects not focused on noise pollution impacts in the noisy Anthropocene. Our findings provide evidence of a significant negative impact of noise on habitat selection and on vocal communication at the site level in songbirds, emphasizing the need to consider noise impacts when planning infrastructure in natural environments. Altogether acoustic monitoring enabled us to detect tnoise-related behavioral impacts and describe the noise patterns responsible. This dual-purpose approach positions automated recording and signal detection as essential for bridging monitoring innovations with evidence-based conservation action.